Treball de Fi de Grau en Física

# Training of Machine Learning Models for Stellar Characterization

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**Abstract**

Reliable stellar masses are required for stellar astrophysics and for the interpretation of exoplanet observations, but direct and homogeneous mass determinations are available only for limited samples. We present a probabilistic machine-learning study of main-sequence stellar mass prediction based on a calibration catalogue assembled mainly from detached eclipsing binaries, asteroseismic stars and interferometric benchmark stars. After duplicate removal, main-sequence selection and quality cuts, the final catalogue contains 767 usable stars with traceable stellar parameters and uncertainties. Bayesian Additive Regression Trees, a two-process heteroscedastic sparse Gaussian Process and a Hierarchical Bayesian Neural Network are trained as base predictors, and their posterior predictions are combined through Bayesian Hierarchical Stacking. For FGK-type stars, using $T_{\rm eff}$, $\rho$ and [Fe/H] as input features, the stacking model reaches MAE $= 0.052\,M_\odot$, MARD $= 4.66\%$ and MRD $= 0.29\%$ on a holdout test set, while reducing some of the mass-dependent residual trends present in the individual models. An ARIEL-motivated experiment using $T_{\rm eff}$, [Fe/H] and $L$, and excluding $\log g$ to reduce direct dependence on stellar evolutionary models, gives BHS masses that agree closely with grid-based estimates, with MAE $= 0.031\,M_\odot$ and MARD $= 2.86\%$. Finally, models trained on asteroseismic stars are applied to eclipsing-binary components to test whether tidal distortion affects the transferred mass predictions. For the selected sample of 85 binaries, oblateness does not produce a significant signed bias, but it is associated with an increase in the absolute mass residuals, indicating a loss of precision rather than a systematic loss of accuracy. Overall, Bayesian Hierarchical Stacking provides competitive stellar mass estimates while exposing the main limitations of the approach: training-set coverage, uncertainty calibration and the transfer of single-star-trained models to distorted binary systems.

*Resumen*

*Las masas estelares fiables son necesarias para la astrofísica estelar y para la interpretación de observaciones de exoplanetas, pero las determinaciones directas y homogéneas de masa sólo están disponibles para muestras limitadas. En este trabajo se presenta un estudio de aprendizaje automático probabilista para la predicción de masas estelares de secuencia principal, basado en un catálogo de calibración construido principalmente a partir de binarias eclipsantes separadas, estrellas astrosísmicas y estrellas de referencia interferométricas. Tras eliminar duplicados, seleccionar estrellas de secuencia principal y aplicar cortes de calidad, el catálogo final contiene 767 estrellas utilizables con parámetros estelares e incertidumbres trazables. Se entrenan Árboles de Regresión Aditiva Bayesiana, un Proceso Gaussiano disperso heterocedástico de dos procesos y una Red Neuronal Bayesiana Jerárquica como predictores de base, y sus predicciones posteriores se combinan mediante Apilamiento Bayesiano Jerárquico. Para estrellas de tipo FGK, usando $T_{\rm eff}$, $\rho$ y [Fe/H] como variables de entrada, el modelo de apilamiento alcanza MAE $= 0.052\,M_\odot$, MARD $= 4.66\%$ y MRD $= 0.29\%$ en un conjunto de prueba reservado, reduciendo a la vez algunas de las tendencias residuales dependientes de la masa presentes en los modelos individuales. Un experimento motivado por ARIEL que usa $T_{\rm eff}$, [Fe/H] y $L$, y excluye $\log g$ para reducir la dependencia directa de los modelos de evolución estelar, produce masas BHS que concuerdan estrechamente con las estimaciones basadas en mallas evolutivas, con MAE $= 0.031\,M_\odot$ y MARD $= 2.86\%$. Finalmente, los modelos entrenados con estrellas astrosísmicas se aplican a componentes de binarias eclipsantes para comprobar si la distorsión mareal afecta a las predicciones de masa transferidas. Para la muestra seleccionada de 85 binarias, el achatamiento no produce un sesgo significativo en los residuos con signo, pero sí se asocia con un aumento de los residuos absolutos de masa, lo que indica una pérdida de precisión más que una pérdida sistemática de exactitud. En conjunto, el Apilamiento Bayesiano Jerárquico proporciona estimaciones competitivas de masas estelares y, al mismo tiempo, expone las principales limitaciones del enfoque: la cobertura del conjunto de entrenamiento, la calibración de incertidumbres y la transferencia de modelos entrenados con estrellas aproximadamente aisladas a sistemas binarios distorsionados.*

# Contents

# 1 Introduction

Stellar characterisation is a central problem in modern astrophysics. Quantities such as stellar mass, radius, effective temperature, luminosity, surface gravity and metallicity determine the physical state and evolutionary stage of a star, and they also affect the interpretation of any planetary system orbiting it. In particular, the properties inferred for exoplanets depend directly on the parameters adopted for their host stars. Reliable stellar parameters are therefore required not only for stellar physics itself, but also for broader applications such as exoplanet characterisation and population studies.

The direct determination of stellar parameters is, however, not always possible. Different observational methods constrain different quantities with different levels of model dependence. Detached eclipsing binaries provide strong dynamical constraints on masses and radii, asteroseismology probes the internal structure of stars through their oscillation frequencies, and interferometry can give empirical radii and effective temperatures for suitable nearby stars. Spectroscopy remains essential for atmospheric quantities such as metallicity, but it also introduces dependencies on stellar-atmosphere models. For this reason, the construction of a useful calibration sample requires combining several catalogues while keeping the provenance, uncertainties and limitations of the labels under control.

The aim of this work is to train and evaluate probabilistic machine-learning models for stellar mass prediction using a carefully assembled catalogue of main-sequence stars. The models considered are Bayesian Additive Regression Trees, a heteroscedastic sparse Gaussian Process, a Hierarchical Bayesian Neural Network and a Bayesian Hierarchical Stacking model that combines the predictions of the individual methods. The Bayesian framework is used because the problem is not limited to obtaining point estimates: the uncertainty of the predictions is also relevant, especially when the models are applied to regions of parameter space that are sparsely represented in the training data.

The first part of the analysis focuses on mass predictions for FGK-type stars. These stars are well represented in the compiled catalogue and are common targets in stellar and exoplanet studies. The models are trained using stellar parameters such as $T_{\rm eff}$, $\rho$ and [Fe/H], and their performance is evaluated on holdout samples through metrics such as the Mean Absolute Error, the Mean Absolute Relative Difference and the Mean Relative Difference. This provides a baseline assessment of the predictive accuracy and bias of each method, as well as of the improvement obtained through Bayesian Hierarchical Stacking.

A second application is connected with the ARIEL stellar-characterisation effort. In this case, the objective is to produce data-driven stellar-mass estimates that are as independent as possible from stellar evolutionary models. For this reason, the models are trained using $T_{\rm eff}$, [Fe/H] and $L$, avoiding $\log g$ as an input because of its stronger dependence on stellar-model assumptions. The resulting predictions are compared with masses obtained from stellar evolutionary grids and with masses derived from previously published linear relations, providing a benchmark against both model-based and empirical approaches.

Finally, the work studies whether masses predicted for eclipsing-binary components are affected by binary interaction. This is motivated by the use of FGK-type primaries in eclipsing binaries to obtain information about their M-type companions, which are intrinsically difficult to characterise. The relevant question is whether predictions trained on approximately single-star samples can be safely transferred to stars in binary systems. To address this, an oblateness parameter is computed for eclipsing-binary components and the mass residuals are analysed as a function of this quantity. The goal is to distinguish between a systematic prediction bias and a loss of precision associated with tidal distortion.

Overall, this thesis combines catalogue construction, Bayesian machine learning and astrophysical validation tests in order to assess how reliably stellar masses can be inferred from observable stellar parameters. The emphasis is placed not only on predictive performance, but also on uncertainty calibration, model dependence and the physical interpretation of residual trends.

# 2 Machine Learning Models

## 2.1 Bayesian Framework for the Models

Bayesian statistics treats unknown model quantities as uncertain variables. The information available before analysing the data is encoded in a prior distribution; after conditioning on the observations, this information is updated into a posterior distribution. In compact form, Bayes' theorem states that $p(\theta \mid \mathcal{D}) = p(\mathcal{D} \mid \theta)p(\theta)/p(\mathcal{D})$, where $\theta$ represents model parameters, $\mathcal{D}$ the data and $p(\mathcal{D})$ the evidence, which normalizes the posterior distribution. This is useful in scientific modelling because it makes assumptions explicit and allows uncertainty to be carried through the full inference process (McElreath, 2020a).

This framework is especially relevant for stellar characterization. The training data contain observational errors, correlations between stellar parameters and regions of sparse feature coverage in parameter space. In this context, the input stellar quantities used by the models are referred to as features. A Bayesian treatment is therefore not chosen only to obtain point predictions, but to quantify how reliable those predictions are. The models discussed in the following subsections use priors to regularize flexible functions and posterior inference to propagate both data uncertainty and model uncertainty into the final estimates (Chipman et al., 2010; Rasmussen & Williams, 2006; Jospin et al., 2022).

For the models used here, the posterior distribution is usually too complex to evaluate analytically. Markov Chain Monte Carlo (MCMC) methods address this by constructing a sequence of dependent samples whose long-run distribution approximates the target posterior. A proposed move is accepted or rejected according to how compatible it is with the posterior, so the chain spends more time in high-probability regions without requiring an exhaustive search of the full parameter space (Hastings, 1970; Bardenet et al., 2017). However, because consecutive samples are correlated, practical MCMC requires convergence checks, an initial warm-up or burn-in period, and attention to the effective number of independent samples. Although computationally expensive, this sampling approach is well suited to the Bayesian models in this work because it provides a principled way to explore parameter uncertainty rather than collapsing the model to a single point estimation without error.

Rather than selecting a single best model, this work combines the predictions of independent level-0 models through Bayesian Hierarchical Stacking (BHS), as sketched in Figure 1.

In the following sections, a Bayesian Additive Regression Trees, a Gaussian Process and a hierarchical Bayesian neural network model are described. The predictions yielded by these models can be passed to this first-level model, which assigns feature-dependent weights to each one, in order to optimize predictive performance.

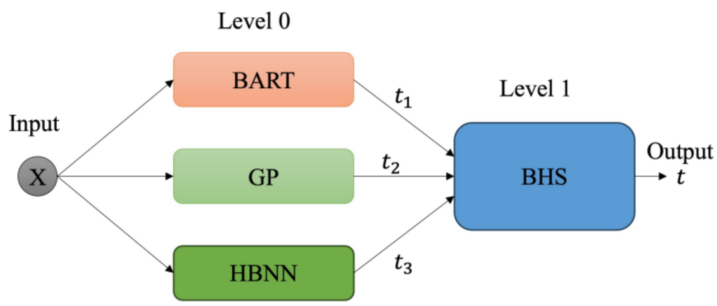



**Figure 1:** Schematic architecture of the Bayesian Hierarchical Stacking model. The predictions from BART, GP and HBNN are combined through a first-level model that learns their relative weights.

## 2.2 Bayesian Additive Regression Trees (BART)

Among the three level-0 models used in this work, the practical deployment of the BART framework is mathematically and computationally the most straightforward, since its architecture requires relatively few tuning choices (Hill et al., 2020). To understand the ensemble, the mechanics of a single regression tree must be defined. A tree recursively partitions the dataset by evaluating binary thresholds across input variables (e.g. $T_{\text{eff}} < 5000$ K). At each step, the algorithm selects the threshold that maximizes the homogeneity of the target variable in the resulting subsets. When predicting for a new observation, the input traverses this learned path of decisions until it reaches a terminal node, or "leaf". The target variable prediction associated with that leaf is simply the mean value of the training points that occupy that specific terminal node.

The Bayesian Additive Regression Trees framework, introduced by Chipman et al. (2010), employs a sum-of-trees architecture to approximate complex non-linear functions. A Random Forest constructs an ensemble of deep, independent regression trees. By training each one of such trees on a random sample of the dataset and averaging the predictions of all of them, it produces a final output which reduces variance. BART follows this dataset subsampling logic, but the final predictions are not the average of the independent models, but the cumulative sum of the terminal node values across all interdependent trees in the ensemble. Therefore, in Random Forests every tree is a strong learner which produces an absolute value prediction for the target; while in BART the trees are weak learners, which just produce a part of the answer (Chipman et al., 2010).

This weak learner approach is mathematically guaranteed by regularisation priors, which constrain the size and structural influence of each tree. These mathematical penalties prevent any single tree component from dominating the overall model, thereby preserving the functional and computational advantages of the additive representation. Left unconstrained, the algorithm naturally drives individual trees to grow massive in pursuit of tighter data fits

(Chipman et al., 2010). The priors counteract this by forcing each tree to function as a weak learner, modelling only a restricted subset of the underlying function. This way, each tree provides only a small part of the predictor space, not a prediction itself.

Since each individual tree recursively partitions the predictor space utilising binary decision rules, the overall capacity and complexity of the model depends on the total number of trees. Predictive performance generally improves as this quantity increases, eventually reaching an asymptote.

Model optimisation follows the Bayesian backfitting MCMC scheme introduced in Section 2.1. During each sweep, the algorithm updates one tree conditional on the partial residual left by the rest of the ensemble, proposing local changes such as growing, pruning or modifying a split. Repeating this process samples plausible tree structures and terminal-node values rather than selecting a fixed ensemble (Chipman et al., 2010).

Crucially for stellar characterization, where uncertainty of the input features is a dominant factor, the framework used in this work accommodates an errors-in-variables approach. Inputs are not treated as static, exact measurements; rather, they are modeled as latent variables characterised by a Gaussian distribution whose variance is fixed by the observational uncertainties. This allows the model to account for the diverse uncertainties present across the entire star catalogue.

## 2.3 Two-process Heteroscedastic Sparse Gaussian Process (GP)

In classical parametric modelling, the functional form of the predictor is fixed in advance and inference is performed over its parameters. A Gaussian Process (GP) takes a different approach: it defines a probability distribution over possible functions. Any finite set of function values is assumed to follow a joint Gaussian distribution, so the model is specified by a mean function and a covariance function, or kernel. In many applications, the covariance function carries most of the information, while the mean function can be set to zero. Therefore, the main modelling choices are the prior covariance structure and the observational noise model, rather than an explicit fixed equation for the underlying function (MacKay, 1998; Rasmussen & Williams, 2006).

Training a GP therefore amounts to selecting a kernel and inferring its constraints, usually referred to as hyperparameters. The amplitude of the kernel controls the overall scale of the functions, and its length-scale controls how rapidly function values can change along the input axes. In this work, the kernel is chosen to favour smooth functions: nearby points in the input space are strongly correlated, while the correlation decreases as their separation increases. Large length-scales are thus dominant, since they produce slowly varying functions (Rasmussen & Williams, 2006).

To prevent the model from choosing unphysical hyperparameter values, priors are placed on the kernel length-scales and amplitudes. The version used in this work is the Automatic Relevance Determination (ARD) kernel, where each input dimension has its own length-scale, rather than sharing a single one. Dimensions with little predictive information can therefore be assigned very large length-scales, making them contribute less to the covariance structure

(Rasmussen & Williams, 2006). The priors regularise this process by keeping the length-scales compatible with the spacing and span of the training data, while weakly informative amplitude priors avoid variances that collapse to zero or diverge.

The standard GP formulation usually assumes homoscedastic observational noise, meaning a single constant noise variance across the dataset. Stellar catalogues, however, contain measurements with uncertainties that vary between stars and between parameters. To account for this, the model used in this work follows a two-process heteroscedastic construction: one GP, $f(\boldsymbol{x})$, represents the latent predictive mean as a function of the stellar parameters, while a second GP, $g(\boldsymbol{\epsilon})$, models the logarithm of the noise variance from the reported measurement errors. This allows the predictive uncertainty to vary across the input space instead of being forced into a single global noise level (Le et al., 2005).

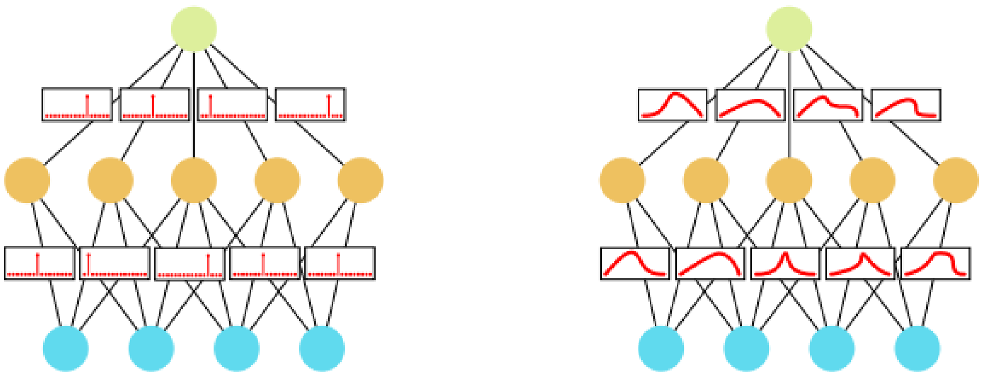
This flexibility comes with a computational cost. Because the kernel compares each data point with every other one, standard GP inference constructs an $N \times N$ covariance matrix; its diagonal terms describe each point's variance, while the off-diagonal terms encode similarity between different observations (Rasmussen & Williams, 2006). GP inference scales as $\mathcal{O}(N^3)$ because it requires operations on the full covariance matrix. To make inference feasible, both processes $f(\boldsymbol{x})$ and $g(\boldsymbol{\epsilon})$ are represented through a sparse approximation with $M$ inducing points, which act as representatives of the entire training set. Predictive covariances are therefore computed through the inducing set rather than directly through all pairwise relations between the $N$ observations, reducing the computational complexity to approximately $\mathcal{O}(NM^2)$ (Quiñonero-Candela & Rasmussen, 2005). Even so, Gaussian Processes can remain computationally demanding for large datasets.

The model is treated in a fully Bayesian way. As described in Section 2.1, MCMC sampling explores plausible kernel hyperparameters, inducing variables and noise structures rather than fixing them at a single point with no uncertainty. Predictions are then averaged over these samples, propagating measurement and model uncertainty into the reported intervals.

## 2.4 Hierarchical Bayesian Neural Network (HBNN)

At its core, an artificial neural network is a sequence of mathematical operations arranged in layers. Neural networks are used to approximate non-linear relations by applying successive linear transformations and non-linear activation functions to the input data. In a standard network, the weights and biases in each layer are fitted as fixed point estimates. This can provide accurate predictions, but it does not naturally represent uncertainty in the model parameters and may produce overconfident results outside the region well covered by the training data (Jospin et al., 2022).

Bayesian Neural Networks (BNNs) address this limitation by assigning probability distributions to the weights and biases instead of single values. After conditioning on the data, the prediction is obtained by averaging over an ensemble of possible networks, making model uncertainty part of the prediction itself (Blundell et al., 2015). In the hierarchical formulation used in this work, the model also accounts for the finite precision of the observations, not only for uncertainty in the neural-network weights and biases (Jospin et al., 2022). This distinction is central for stellar characterization, where sparse training regions and measurement errors both affect the reliability of the final prediction.



(a) Point-estimate NN.  (b) BNN with weight distributions.

**Figure 2:** Comparison between neural-network formulations. Figure 2a shows a standard network, in which the weights and biases are fixed after training. Figure 2b shows the Bayesian formulation used in this work, where probability distributions are assigned to the weights and biases. Adapted from (Jospin et al., 2022).

The network itself uses Gaussian priors for the weights and biases and a two-layer architecture with Leaky ReLU activations (Maas et al., 2013). As discussed in Section 2.1, MCMC sampling explores the posterior distribution rather than converging to a single point with no uncertainty. The HBNN therefore keeps the flexibility of neural networks while preserving the probabilistic interpretation required for calibrated stellar predictions (Jospin et al., 2022).

## 2.5 Bayesian Hierarchical Stacking

The predictions from BART, GP and HBNN are combined through Bayesian Hierarchical Stacking (BHS). Stacking is an ensemble method in which the outputs from several models are combined using learned weights, allowing relevant information from each model to contribute to the final prediction (Wolpert, 1992). Unlike simple averaging, BHS allows the weights assigned to each level-0 predictor to depend on the input features. This is important because a model may be reliable for near-solar stars but less reliable for more massive objects or for stars with larger uncertainties. The former is the case of the level-0 models used in this work, as described in Section 5. The final estimate is therefore a locally weighted combination of the three model outputs (Yao et al., 2022).

The weights are estimated using cross-validation information, discussed in Section 2.6. The hierarchical formulation regularizes the variation in these weights, preventing the stacking model from reacting too strongly to noise in sparsely populated regions of the dataset. Thus, BHS is used as a controlled way to exploit the complementary strengths of the three models while reducing the risk of relying on a single imperfect predictor.

## 2.6 Model Evaluation through Cross-validation

Model evaluation estimates how well a trained model will generalize to stars that were not used during training. A simple train–validation split gives a direct test of this: part of the dataset is kept aside as a validation subset, and the trained model is then evaluated against it. Nevertheless, the result can depend strongly on the particular partition, since a large validation set reduces the data available for fitting, while a small one gives a noisy estimate of the error (Rasmussen & Williams, 2006).

Cross-validation reduces this dependence on a single split. In $k$-fold cross-validation, the dataset is divided into $k$ subsets; each subset is used once for validation while the remaining subsets are used for training. The reported performance is then averaged over the folds, so every star contributes both to fitting and to out-of-sample assessment (Rasmussen & Williams, 2006). This provides a more stable basis for evaluating the target-variable predictions produced by the models used in this work, allowing for a better comparison between them.

The most exhaustive version is Leave-One-Out Cross-Validation (LOO-CV), where each observation is validated once while the model is trained on all remaining observations. For complex Bayesian models, explicitly refitting the model for every observation is usually too expensive. Therefore, when appropriate, this work uses the efficient LOO approximation proposed by Vehtari et al. (2017): Pareto-smoothed importance sampling (PSIS-LOO). This gives a common evaluation criterion without requiring a separate validation set.

# 3 Methods

This section introduces the observational basis of the stellar labels used later to construct the final catalogue. The catalogue is built from three reference methods: detached eclipsing binaries, asteroseismology and interferometry. Spectroscopy is discussed first as context, because it is essential for atmospheric quantities, especially metallicity, but relies strongly on stellar-atmosphere modelling. This contrast motivates the use of the other three methods as the main sources of high-quality labels, while also making clear that they are not completely free from auxiliary assumptions (Serenelli et al., 2021).

## 3.1 Spectroscopy

Spectroscopy analyses the stellar radiation received as a function of wavelength. The continuum, discontinuities, absorption lines and molecular bands are produced by physical processes in the stellar atmosphere, and therefore contain information about quantities such as $T_{\text{eff}}$, $\log g$ and [Fe/H].

The difficulty is that these quantities are not usually obtained by direct measurement. They are inferred by comparing the observed spectrum with synthetic spectra, line lists and stellar-atmosphere models (REFERENCE NEEDED). Spectroscopy is therefore essential, especially for metallicity, but it can also introduce model-dependent assumptions into any machine-learning model trained on those labels.

## 3.2 Detached Eclipsing Binaries

Detached eclipsing binaries are systems in which two stars orbit each other without significant mass transfer, while their orbital plane causes periodic eclipses as seen from Earth. Because the components are dynamically linked but remain physically separated, they are especially valuable for obtaining fundamental stellar parameters (Southworth, 2015; Eker et al., 2018).

The light curve gives the orbital period, eclipse geometry, relative radii and temperature ratio. Radial-velocity measurements from Doppler shifts then provide the orbital velocities. Combining these observables with Keplerian dynamics yields absolute masses and radii, from which density and surface gravity follow. These quantities are therefore mainly geometric and dynamical.

The atmospheric quantities are less direct. Absolute temperatures and luminosities usually require either spectroscopy, colour-temperature relations, parallax and bolometric fluxes, or a combination of these ingredients (Graczyk et al., 2022). For this reason, detached eclipsing binaries are treated as a strong reference source for masses and radii, while their temperature and luminosity labels still require attention to the assumptions used to derive them.

## 3.3 Asteroseismology

Asteroseismology studies stellar oscillations. In solar-like stars, turbulent motion in the outer convective envelope excites pressure waves that propagate through the interior and produce small brightness variations. Time-series photometry and Fourier analysis reveal the characteristic oscillation frequencies of the star (Bellinger et al., 2019; Serenelli et al., 2021).

Two global seismic quantities are especially useful. The large frequency separation, $\Delta\nu$, scales with the square root of the mean density, while the frequency of maximum oscillation power, $\nu_{\max}$, is related to surface gravity and effective temperature:

$$\Delta\nu \propto \sqrt{\frac{M}{R^3}}, \qquad \nu_{\max} \propto \frac{M}{R^2\sqrt{T_{\text{eff}}}}.$$

With an external estimate of $T_{\text{eff}}$, these relations constrain mass, radius and $\log g$ (REFERENCE NEEDED).

This makes asteroseismology particularly useful for single stars, where direct dynamical masses are usually unavailable. However, the method still depends on scaling relations, external atmospheric inputs and, in more detailed analyses, stellar models. It therefore provides strong constraints, but not completely model-independent labels.

## 3.4 Interferometry

Interferometry combines the light collected by separated telescopes, effectively increasing the angular resolution beyond that of a single aperture. For nearby bright stars, this makes it possible to measure the angular diameter. Together with an independent distance, usually from parallax, the stellar radius can then be obtained geometrically (REFERENCE NEEDED).

If the bolometric flux is also known, the effective temperature follows from the Stefan-Boltzmann relation. This makes interferometry one of the most empirical routes to stellar radii and temperatures, and it is used in benchmark-star calibrations (Soubiran et al., 2024). The method is not fully assumption-free, because bolometric flux estimates require extinction corrections and model-dependent treatment of unobserved spectral regions. It is also limited to stars with sufficiently large angular diameters, so its contribution to the total sample is small.

### 3.5 Limits of Model Independence

The methods described above are used because they reduce the dependence on stellar-atmosphere models, but they do not remove modelling assumptions completely. Metallicity is the clearest case: [Fe/H] or [M/H] must be inferred from spectra through line-formation calculations, synthetic spectra and atmospheric models (REFERENCE NEEDED). Detached eclipsing binaries still require external calibrations for absolute temperatures and luminosities (Graczyk et al., 2022). Asteroseismic masses and radii depend on scaling relations and an external $T_{\rm eff}$ estimate, while interferometric temperatures require bolometric fluxes and extinction corrections (Serenelli et al., 2021; Soubiran et al., 2024).

For this reason, the catalogues used in this work should not be interpreted as perfectly model-independent ground truth. They are instead high-quality reference labels, selected because their central constraints are closer to geometry or dynamics than to atmospheric modelling: eclipsing binaries provide strong masses and radii, asteroseismology constrains stellar density and surface gravity, and interferometry gives empirical radii and temperatures for suitable nearby stars. This compromise is important for the machine-learning models trained later, since their predictions cannot be more physically reliable than the labels from which they learn.

## 4 Data

In supervised machine learning, the quality of the labels can be as important as the size of the training set. Large catalogues affected by heterogeneous systematics may lead the models to reproduce those biases instead of learning the underlying physical relations (Li & Chao, 2021). For this reason, this work follows a quality-centred strategy, prioritising a smaller, traceable and physically reliable calibration set over a larger but less controlled sample.

Following the guidance of Moya et al. (2018), an extended stellar catalogue was assembled from literature sources with complementary parameter coverage. The aim was to obtain a statistically meaningful and heterogeneous sample, while keeping the provenance and uncertainties of the stellar parameters as clear as possible.

As explained in Section 3, the reference sources were selected around three observational methods: detached eclipsing binaries, asteroseismology and interferometry. Studies published between 2018 and 2025 were reviewed when they provided uncertainties and as many of the required stellar parameters as possible, listed in Table 1. Stellar mass and radius were mandatory, since stellar densities were computed from them. This initial compilation contained 16094 stars, before duplicate removal and quality cleaning.

## 4.1 Catalogues

The literature sources used to compile the complete data sample are briefly described below. Eker et al. (2018) provided accurately measured components of detached eclipsing spectroscopic binaries, used to calibrate $L$ and $T_{\text{eff}}$ for main-sequence stars over a broad mass range. Hełminiak et al. (2019) is also based on detached eclipsing binaries, but includes spectroscopic support; these entries therefore required additional care. Its reported chemical information was given as $[\alpha/\text{Fe}]$ and was not directly usable in the metallicity scale adopted here.

Graczyk et al. (2021) and Graczyk et al. (2022) used detached eclipsing binaries to calibrate surface-brightness–colour relations; the latter work provides the dwarf-star calibrating sample adopted in this catalogue. Nsamba et al. (2025) presents a collection of pre-existing benchmark stars, whose overlap with other samples required careful duplicate removal. Bellinger et al. (2019) provided asteroseismic masses and radii for mostly solar-like stars, although several objects lacked some of the atmospheric parameters required in this work.

Serenelli et al. (2021) is a broad review and benchmark compilation of stellar mass-determination methods across the HR diagram, including detached eclipsing binaries and asteroseismology. It contributed a large number of reliable stars, although luminosity and metallicity were not always available. Soubiran et al. (2024) provided the third version of the Gaia FGK benchmark-star catalogue, with fundamental $T_{\text{eff}}$ and $\log g$ values derived from angular diameters, bolometric fluxes, parallaxes and masses, but without luminosities in the form required here. Xiong et al. (2023) constructed an empirical library from eclipsing binaries while including some spectroscopic information; because of the non-standard identifiers and some comparatively large uncertainties, these entries were treated with additional caution during cross-matching.

Sayeed et al. (2025) presented a homogeneous catalogue of Kepler main-sequence and subgiant solar-like oscillators, deriving global asteroseismic parameters, although some quantities, such as metallicity, were not accompanied by uncertainties. Finally, Yıldız et al. (2019) derived masses and radii for Kepler and CoRoT solar-like oscillators using asteroseismic frequency diagnostics. When metallicities were provided as surface metal mass fractions $Z_s$, they were converted to the logarithmic scale using $[\text{M/H}] = \log_{10}[(Z_s/X_s)/(Z/X)_\odot]$, adopting $X_s \simeq 0.7381$ and $(Z/X)_\odot = 0.0181$ from Asplund et al. (2009).

## 4.2 Cleaning

When the same star appeared in several catalogues, the retained entry was chosen according to the reliability of the source. Priority was given to catalogues providing the most direct stellar parameters, smaller reported uncertainties and better parameter completeness. Regularly updated catalogues and benchmark-star compilations were also preferred over more heterogeneous or less traceable sources. In practice, this meant favouring high-precision eclipsing-binary catalogues such as Eker et al. (2018) and the calibrating samples of Graczyk et al. (2022), benchmark compilations such as the Gaia FGK benchmark-star catalogue (Soubiran et al., 2024), and samples cross-validated across different methods, such as those in Serenelli et al. (2021). After duplicate removal, 13783 stars remained.

The cleaned sample was then restricted to main-sequence candidates using a criterion based on $\log g$. Following Bilir et al. (2008), stars with $\log g > 4.0$ were classified as main-sequence objects. Surface gravities in this range are characteristic of dwarf stars, while lower values are generally associated with more evolved objects, such as subgiants, post-main-sequence stars and red giants. Extremely large values of $\log g$ ($\gtrsim 5.5$) were not interpreted as normal main-sequence stars.

Although this threshold is approximate, the effect of applying different classification criteria was already analysed by Moya et al. (2018). They found that the choice between a simple threshold and a more robust classification method had little influence on the final results. The $\log g$ criterion adopted here was therefore considered sufficient for the purposes of this work.

After this selection, 1915 stars remained. Not all of them contained every required parameter. In principle, incomplete entries could be retained in a Bayesian framework by treating missing quantities as latent variables, provided that an explicit model for the missing-data process is specified (McElreath, 2020b, p. 511). In this work, however, parameter completeness and direct traceability were prioritised, so this kind of imputation was not used.

The 1915-star sample was then cross-matched with the Moya 2022 catalogue in order to merge both sources, avoid duplicates and retain only the highest-quality entries. Following the same data-centred strategy, stars were removed when they had excessively large uncertainties, zero reported uncertainty in any parameter except $a$, were not classified as main-sequence stars in both catalogues, or had duplicated identifiers. This reduced the sample to 784 stars. Section 5.1 explains the later removal of 17 additional objects, leading to the final training sample of $n = 767$ stars.

**Table 1.** Summary of the stellar parameters contributed by each catalogue to the final calibration sample of 767 usable main-sequence stars. Each value gives the number of stars with the corresponding available parameter. The metallicity column combines [Fe/H] and [M/H]. Catalogues not individually discussed in this work are described in Moya et al. (2018).

| Catalogue | $M$ ($M_\odot$) | $R$ ($R_\odot$) | $T_{\rm eff}$ (K) | $L$ ($L_\odot$) | Meta (dex) | $\log g$ (cm s$^{-2}$) | $\rho$ ($\rho_\odot$) | $a$ ($R_\odot$) |
|---|---|---|---|---|---|---|---|---|
| Serenelli et al. (2017) | 240 | 240 | 240 | 240 | 240 | 240 | 240 | 0 |
| Xiong et al. (2023) | 141 | 141 | 141 | 141 | 141 | 141 | 141 | 141 |
| Chaplin et al. (2014, SDSS) | 62 | 62 | 62 | 62 | 62 | 62 | 62 | 0 |
| Eker et al. (2014) | 54 | 54 | 54 | 54 | 54 | 54 | 54 | 54 |
| Yildiz et al. (2019) | 42 | 42 | 42 | 42 | 42 | 42 | 42 | 0 |
| Lund et al. (2016) | 28 | 28 | 28 | 28 | 28 | 28 | 28 | 0 |
| Huber et al. (2013) | 24 | 24 | 24 | 24 | 24 | 0 | 24 | 0 |
| Silva Aguirre et al. (2015) | 24 | 24 | 24 | 24 | 24 | 24 | 24 | 0 |
| Wang et al. (2025) | 20 | 20 | 20 | 20 | 20 | 20 | 20 | 20 |
| Eker et al. (2014, 2018) | 17 | 17 | 17 | 17 | 17 | 17 | 17 | 17 |
| Silva Aguirre et al. (2017) | 15 | 15 | 15 | 15 | 15 | 15 | 15 | 0 |
| Soubiran et al. (2024) | 13 | 13 | 13 | 13 | 13 | 13 | 13 | 0 |
| Helminiak et al. (2019) | 11 | 11 | 11 | 11 | 11 | 11 | 11 | 11 |
| Pan et al. (2025) | 10 | 10 | 10 | 10 | 10 | 10 | 10 | 10 |
| Ligi et al. (2016) | 10 | 10 | 10 | 10 | 10 | 0 | 10 | 0 |
| Chaplin et al. (2014, Bruntt) | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 0 |
| Özdarcan et al. (2025, Feb.) | 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 |
| Malkov et al. (2007) | 4 | 4 | 4 | 4 | 4 | 0 | 4 | 0 |
| Welsh et al. (2012) | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 |
| Brogaard et al. (2018) | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 |
| Özdarcan et al. (2024) | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 |
| Torres et al. (2010) | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 |
| Çakırlı et al. (2025, Apr.) | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 |
| Çakırlı et al. (2025, Dec.) | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 |
| Knudstrup et al. (2020) | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 |
| Martin et al. (2024, Dec.) | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 |
| Miller et al. (2022) | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 |
| Miller et al. (2026) | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 |
| O'Brien et al. (2025) | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 |
| Xu et al. (2026) | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 |
| Yücel et al. (2026) | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 |
| Bond et al. (2020) | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| Jennings et al. (2024) | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 |
| Lund et al. (2019) | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 |
| Maxted et al. (2023) | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| Maxted et al. (2025) | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| Thomsen et al. (2022) | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| Total | 767 | 767 | 767 | 767 | 767 | 729 | 767 | 295 |

Although each star in the final merged sample is assigned a primary catalogue identifier according to the reliability criteria described above, the individual stellar parameters for a given entry do not always originate from that single source. To maximise completeness, the merging algorithm constructed composite entries: when the primary catalogue lacked a given parameter, the missing value was filled from an available secondary catalogue. The final parameter set for a star can therefore be heterogeneous in origin. To preserve traceability, a more detailed internal provenance table was used alongside Table 1; for each primary source, it records which secondary catalogues were queried to fill each missing parameter.

When several secondary sources were available, they were prioritised according to the same reliability hierarchy.

The Hertzsprung–Russell diagram was used as a final consistency check for the cleaned sample. This diagram relates stellar luminosity to effective temperature and is commonly used to identify evolutionary stages, especially the main sequence. As shown in Figure 3, the stars mostly follow the expected main-sequence behaviour, with luminosity decreasing toward lower effective temperatures. FGK stars occupy approximately $T_{\rm eff} \sim 3500$–$7400$ K (Pecaut & Mamajek, 2013), or $\log_{10} T_{\rm eff} \sim 3.54$–$3.87$. Most of the sample lies in this regime, especially the asteroseismic and interferometric stars. The eclipsing-binary subsample is also mainly concentrated around the main sequence, but spans a broader region of the diagram, including both hotter and more luminous objects and cooler, less luminous ones.

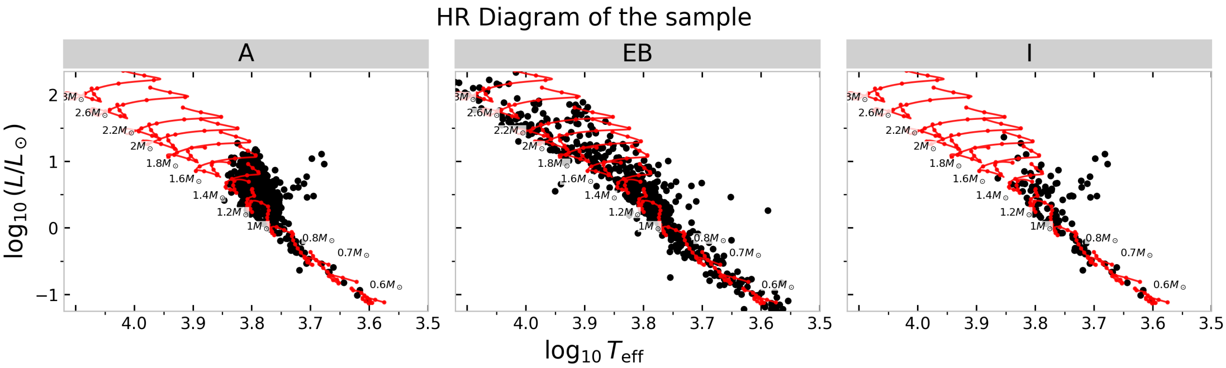



**Figure 3:** HR diagram of the 767 stars in the final sample. The red curves are theoretical evolutionary tracks of different masses obtained with PARSEC (Bressan et al., 2012). The sample is divided by the main method used to study each star: asteroseismology, eclipsing binaries and interferometry.

Finally, Section 5.3 requires the computation of oblateness (Eq. 2), which depends on the orbital semimajor axis $a$. This parameter was often reported without an uncertainty. To retain a statistically meaningful sample, the missing uncertainty was estimated from four significant digits of the reported value. This produced 295 stars with computed oblateness, 48 of which used this uncertainty estimate.

# 5 Machine Learning Model Training

## 5.1 Mass Predictions for FGK-type Stars

Main-sequence stars of spectral types F, G and K are especially useful in stellar studies because they are common survey targets and therefore often have well-measured atmospheric parameters. By contrast, main-sequence M-type stars are less extensively documented. They are cooler, low-mass objects that burn hydrogen slowly and evolve very little over the age of the Universe (Laughlin et al., 1997). Their intrinsic faintness and slow evolution make precise stellar parameters difficult to obtain (Chabrier & Baraffe, 2000).

Maxted (2023) proposed a way to obtain information about M-type stars by studying eclipsing-binary systems in which the M star is the secondary component of a better-characterized primary star. Because many eclipsing-binary parameters are measured relative to the two components, information about one star can help constrain the other (Southworth, 2015). This motivates a focus on systems with an FGK-type primary and an M-type secondary: accurate mass predictions for the better-characterized FGK component can improve the inferred properties of the lower-mass companion. This helps to extend the calibration towards the M-star regime, where high-quality masses remain scarce.

As discussed in Section 4.2, Figure 3 shows that the catalogue compiled in this work is rich in FGK-type stars, making this first mass-prediction experiment well suited to the available data. Following Moya et al. (2018), $T_{\rm eff}$, $\rho$ and [Fe/H] were adopted as input parameters for stellar-mass estimation. This choice is motivated by well-studied eclipsing binaries, for which these quantities can be obtained more precisely than alternatives such as $\log g$ or $L$. After combining the stars compiled in this work with the Moya catalogue, applying the main-sequence selection described in Section 4, requiring these three input parameters, and removing stars with zero reported uncertainty, the final available set contained 767 stars. Of these, 618 were used for training and the remaining 155 stars, corresponding to 20%, were retained as a holdout set for testing. The holdout set was repeatedly randomly selected for each seed run.

## Results

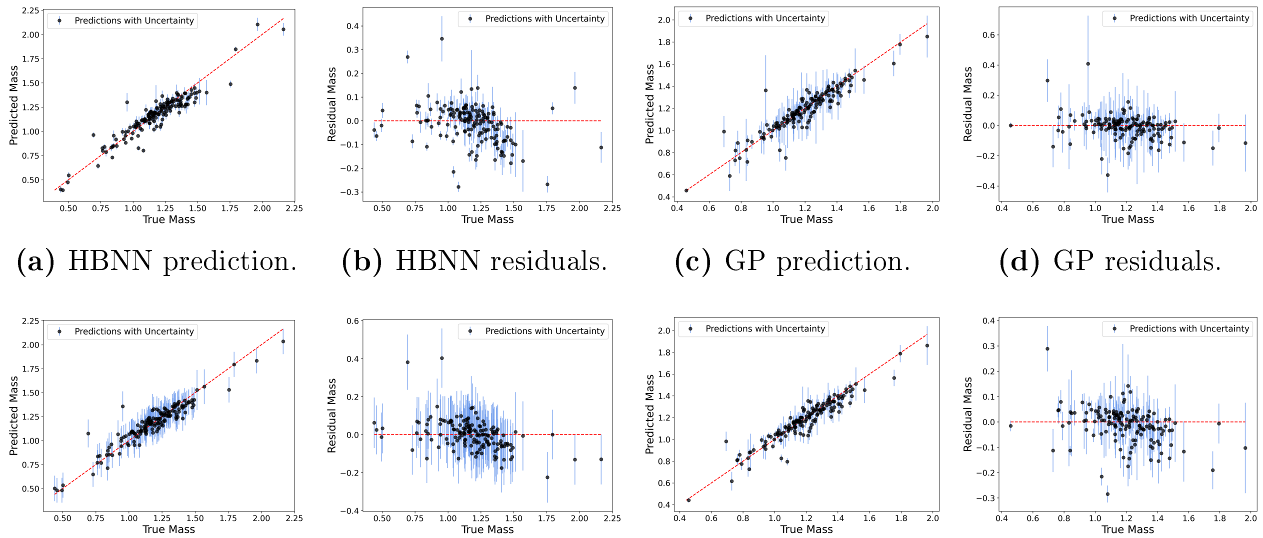
The behaviour described in this section was consistent across different random-seed runs for the four analysed models. The main evaluation metrics are the Mean Absolute Error (MAE), which measures the average absolute difference between the true masses $m_i$ and the model estimates $\hat{m}_i$; the Mean Absolute Relative Difference (MARD), which measures the average relative accuracy across $N$ samples; and the Mean Relative Difference (MRD), which measures the signed relative bias of the predictions with respect to the true values.

$$\mathrm{MAE} = \frac{\sum_{i=1}^{N} |m_i - \hat{m}_i|}{N}, \quad \mathrm{MARD} = \frac{\sum_{i=1}^{N} \frac{|m_i - \hat{m}_i|}{m_i}}{N}, \quad \mathrm{MRD} = \frac{\sum_{i=1}^{N} \frac{m_i - \hat{m}_i}{m_i}}{N}. \tag{1}$$

Figure 4a shows the HBNN mass predictions for one representative run. In the prediction panel, the points mostly follow the 1:1 relation between predicted and true mass. However, the distribution is not completely uniform: lower-mass stars tend to be overpredicted, whereas higher-mass stars tend to be underpredicted. When averaged over the full holdout set, these two effects partly cancel, leading to a Mean Relative Difference close to zero, 0.71%, as shown in Table 2.

This trend is more clearly visible in the residual plot[1], Figure 4b. The residuals change sign at approximately 1.3 $M_\odot$: lower-mass stars tend to have positive residuals, while higher-mass stars tend to have negative residuals. Thus, HBNN performs well for the main, near-solar-mass population, where most predictions agree with $M_{\rm true}$ within $1\sigma$. The model should, however, be treated with caution outside this central region, especially for higher masses.

[1] For visual clarity, residuals were defined as $M_{\rm pred} - M_{\rm true}$ in the figures, including 4, while in the MAE, MARD and MRD definitions the sign is switched, $M_{\rm true} - M_{\rm pred}$.




**(a)** HBNN prediction. **(b)** HBNN residuals. **(c)** GP prediction. **(d)** GP residuals.

**(e)** BART prediction. **(f)** BART residuals. **(g)** BHS prediction. **(h)** BHS residuals.

**Figure 4:** Holdout-test performance of the HBNN, GP, BART and BHS mass models trained with $T_{\rm eff}$, $\rho$ and [Fe/H]. For each model, the prediction panel shows $M_{\rm pred}$ as a function of $M_{\rm true}$, while the residual panel shows $M_{\rm pred} - M_{\rm true}$ as a function of $M_{\rm true}$. Vertical error bars represent the predictive uncertainty returned by each model. The GP and BHS panels are restricted to the stars below the 95th percentile of predicted mass uncertainty in order to keep the main population visually readable. The full calibration sample contains $n = 767$ stars, with 20% reserved for the holdout test.

A plausible explanation is that the models were trained on a sample strongly concentrated around $1M_\odot$. As a result, HBNN tends to pull predictions towards this well-populated region, producing masses closer to the central training distribution than to the true values at the edges of the mass range. This behaviour is expected for Bayesian neural networks when the test inputs differ from the training distribution (Izmailov et al., 2021).

The trend is less clear for the Gaussian Process model, as shown in Figure 4c. For visualization, the GP plots were restricted to stars with predicted mass uncertainty below the 95th percentile, because the full GP output contains a small number of objects with extremely large predicted uncertainties. Including those points compresses the vertical scale and makes the behaviour of the bulk of the sample unreadable.

Most GP predictions remain close to the 1:1 relation, and almost all points are compatible with the reference masses within $1\sigma$. The residual plot in Figure 4d shows a more compact and symmetric distribution around zero than the HBNN residuals in Figure 4b. This is also reflected in Table 2: the GP reaches a MARD of 4.79%, lower than the HBNN value of 5.27%. In addition, the GP has an MRD close to zero, 0.34%, indicating no clear global tendency to underpredict or overpredict stellar masses.

It is, however, notable that the GP uncertainties are larger than in the HBNN case, indicating that the model is underconfident in part of the sample. This remains true even after restricting the plotted set to the stars below the 95th percentile of predicted mass uncertainty. Stars with input-feature uncertainties larger than 9% generated predicted mass uncertainties as large as $10^3\ M_\odot$, so they were removed from the sample, in order to investigate the origin of this extremely underconfident behaviour. This removed 17 objects, but did not eliminate the problem.

In sparse regions of the input space, Gaussian Processes naturally return broader predictive distributions, reflecting extrapolation uncertainty (Cho et al., 2026). This is one of the main advantages of GPs: predictive uncertainty increases when the model has little nearby information. This explains why no strong systematic trend is visible in Figures 4c and 4d. However, the very large uncertainty values found here, sometimes of order $10^2$–$10^3$ $M_\odot$, are not physically meaningful.

A further inspection of the objects with very large predicted uncertainties, after removing those stars with input parameter uncertainties larger than 9%, showed that these input objects were no longer the same across different runs, and no physical pattern was found in their stellar parameters. This suggests that the issue is not only caused by a few erroneous data entries, but is likely connected to instability in the GP posterior or in the variance submodel. Le et al. (2005) discuss GP regression with input-dependent noise and note that directly estimating both the mean and the variance leads to difficult optimisation problems. For this reason, the GP results should be interpreted with caution. The percentile-restricted plots are used only to display the behaviour of the main population, not to hide the existence of these unstable uncertainty estimates.

For BART, Figures 4e and 4f, the result is similar to that obtained with the Bayesian neural-network model: masses below $1M_\odot$ tend to be overpredicted, while masses above $1M_\odot$ tend to be underpredicted. The predictions remain relatively close to the 1:1 relation, yielding a low MARD value of 4.98%. The negative MRD value, $-0.63\%$, shows that low-mass overpredictions and high-mass underpredictions again partially cancel when averaged across the full data interval. Therefore, as in the HBNN case, the MRD alone is not sufficient to describe the model behaviour unless it is interpreted together with the prediction and residual plots.

Here, the reduced accuracy outside the central training region is consistent with a known limitation of tree-based regression models. Standard BART and related tree ensembles are strong interpolators, but their extrapolation ability is limited by their leaf-based, piecewise-constant structure. Predictions outside sparsely sampled regions can become biased towards the dominant range of the training data and may have poorly calibrated uncertainty (Zhang et al., 2019). BART also shows comparatively large uncertainties, so the loss of accuracy away from the central mass range is reflected in underconfident predictions.

Finally, the stacking model was applied to combine the information contained in the individual BART, GP and HBNN predictions. Since both HBNN and BART show systematic behaviour away from the region near $1M_\odot$, the GP model is expected to have a larger influence in those outer regions. Therefore, the BHS plots shown in Figures 4g and 4h are also restricted to stars below the 95th percentile of predicted mass uncertainty, avoiding the visual masking produced by a small number of extremely uncertain points.

Figure 4g shows no clear mass-dependent trend in the BHS predictions. Therefore, the systematic mispredictions observed for BART and HBNN away from one solar mass are not inherited by the stacking model. The resulting MRD value, 0.29%, is close to zero and is the lowest among the four models, supporting the absence of a visible global bias. In this case, the behaviour of the plots confirms that the near-zero MRD is not mainly produced by localized overpredictions and underpredictions cancelling each other out, as happened for HBNN and BART.

Although the GP model clearly influences BHS, since the negative-slope trend is no longer visible, the difference between the BHS MARD, 4.66%, and the GP MARD, 4.79%, shows that the stacking model is not simply mimicking the GP model overall. Instead, it combines the models in a way that improves the final residual metrics while reducing the systematic behaviour seen in some of the individual predictors.

The predictive uncertainties observed in the plots of the BHS model are, however, the largest among the models in this representative run. This is consistent with the fact that stacking combines predictive distributions from all models and can inherit part of the large uncertainties that both the GP and BART components show for some stars (Yao et al., 2022).

**Table 2.** Comparison of the residual metrics obtained by the mass-prediction models on the holdout set. The models were trained with $T_{\rm eff}$, $\rho$ and [Fe/H] as input features, and stellar mass as the target. MAE gives the average absolute error in solar masses, MARD gives the average absolute relative error, and MRD gives the signed relative bias.

| Metric | BART | GP | HBNN | BHS |
|---|---|---|---|---|
| MAE ($M_\odot$) | 0.057 | 0.053 | 0.060 | 0.052 |
| MARD [%] | 4.98 | 4.79 | 5.27 | 4.66 |
| MRD [%] | -0.63 | 0.34 | 0.71 | 0.29 |

The evaluation metrics (Table 2) show that, overall, the relative errors are consistently around the 5% level, which is a strong result for this calibration sample. Similar values were obtained by the BHS model presented by Moya & López-Sastre (2022). However, that model used four input features instead of three, and would therefore have access to more predictive information, as discussed by those authors. The 2022 model did not propagate the observational uncertainties of the stellar parameters. The fact that the present three-feature models reach comparable MARD values while incorporating these uncertainties suggests that the improved probabilistic treatment provides a more robust calibration withour a clear loss of predictive accuracy.

This interpretation is also supported by the MAE comparison. Moya & López-Sastre (2022) reported a stacking-model MAE of 0.048 $M_\odot$, which is very close to the BHS value of 0.052 $M_\odot$ obtained here. Considering the remaining models presented by Moya & López-Sastre (2022), with MAE values in the range 0.06–0.08 $M_\odot$, the models in this work achieve lower absolute errors in all cases, while treating the reported input uncertainties consistently.

Finally, all MRD values are below 1%. For GP and BHS, this is consistent with the absence of a clear global bias in the prediction plots. For HBNN and BART, however, the near-zero MRD should not be interpreted alone: in those cases, it is mainly a consequence of systematic overpredictions and underpredictions cancelling when averaged over the full holdout set.

## 5.2 ARIEL: Model-independent predictions

ARIEL, the Atmospheric Remote-sensing Infrared Exoplanet Large-survey, is ESA's fourth medium-class mission, M4, within the Cosmic Vision programme, and is planned for launch around 2030. It is dedicated to the atmospheric characterization of exoplanets and will study the chemical composition of the atmospheres of approximately 1000 transiting planets. The mission is based on spectroscopic and photometric observations, where the planetary signal is measured relative to the emission of the host star. Accurate and homogeneous stellar characterisation is therefore essential for the interpretation of ARIEL observations.

Within this context, the Age, Mass and Radius sub-working group aims to provide homogeneous estimates of stellar ages, masses and radii for the ARIEL target stars (Ariel Stellar Characterisation Working Group, n.d.). The Astronomy and Astrophysics Department at the Universitat de València manages this stellar characterization working group. The task is approached through complementary implementations, including both stellar-modelling methods and data-driven ML/AI techniques. The former implementation is supported by the Astrophysics and Space Sciences Institute of the Universidade do Porto.

Surface gravity, $g$, is a physically informative quantity for the estimation of stellar masses and radii. Previous empirical and machine-learning studies have shown that its inclusion can improve predictive performance (Moya et al., 2018). However, $g$ is often determined through procedures that depend on stellar models. Using it as an input feature in the data-driven model would therefore reduce the independence between the two complementary approaches considered in the ARIEL stellar-characterization work. For this reason, the models in this section were trained using $T_{\text{eff}}$, [Fe/H] and $L$ as input features. These quantities are not completely free from methodological assumptions, but they avoid the same direct dependence on stellar evolutionary models.

The procedure described in Section 5.1 was then repeated using this updated set of input features, yielding similar results, both visually and quantitatively. In this case, however, an additional comparison was possible. The Institute of Astrophysics of the Universidade do Porto provided a complementary dataset of stellar masses obtained through stellar-modelling methods. These estimates are based on PARSEC, the PAdova and TRieste Stellar Evolution Code, which provides stellar evolutionary tracks and isochrones used to infer stellar properties from theoretical models (Bressan et al., 2012). This dataset was first cleaned to remove stars that had already been used during training, and was then used to obtain mass predictions with the trained models. The new mass predictions, including their uncertainties, were compared with the masses obtained from stellar evolutionary grids. Linear relations developed by Moya & López-Sastre (2022) were added to the comparison as well. This provides a benchmark against both model-based mass estimates and previously published data-driven mass-estimation methods.

### Results

## 5.3 Oblateness and mass predictions of eclipsing binaries

As explained in Section 5.1, one of the aims of this work is to predict the masses of FGK-type components in eclipsing-binary systems, so that information can then be inferred for their M-type companions. This is relevant because M-type stars are intrinsically difficult to characterise: they are cool, low-mass stars with slow evolution, and precise observational constraints are generally harder to obtain. Eclipsing binaries therefore provide one of the most useful routes for obtaining information about these objects. However, this also raises an important question: whether the information inferred from stars in eclipsing-binary systems can be extrapolated to isolated stars of similar type.

Eclipsing-binary components are not completely independent objects. Their mutual gravitational interaction determines their orbital motion, but it can also modify their morphology through tidal forces. In systems with stronger interaction, the stars may become tidally distorted rather than remaining perfectly spherical. This distortion can be quantified through the oblateness parameter $o$, given by equation (43) of Chandrasekhar (1933):

$$o = 0.6446 \cdot \left(1 + 4\frac{m}{m_\star}\right) \cdot \left(\frac{R}{a}\right)^3, \tag{2}$$

where $m_\star$ is the mass of the star whose oblateness is being computed, $m$ is the mass of its companion, $R$ is the stellar radius, and $a$ is the orbital semi-major axis.

To test whether this tidal distortion affects the mass predictions, the models are trained on asteroseismic stars, which are treated here as the single-star reference sample, and then applied to the eclipsing-binary components available in the catalogue. The predicted masses are compared with the reference masses of the eclipsing-binary stars, and the resulting residuals are analysed as a function of oblateness. This follows the general approach of Maxted (2023), where eclipsing binaries are used to evaluate how well single-star relations can be transferred to binary systems.

Since the goal in this section is to obtain the most accurate possible mass predictions before analysing the residual dependence on oblateness, four input features are used, following the recommendation of Moya & López-Sastre (2022): $T_{\rm eff}$, $\log g$, $L$ and [Fe/H].

### Results: holdout test

The models were trained using the asteroseismic stars in the dataset, listed in Table 1. This subset contains 420 stars, of which 154 were retained as the holdout set. Before analysing the residuals as a function of oblateness, it is useful to evaluate the performance of the models on this holdout set. Two competing effects are present. On the one hand, the models now use four input features instead of three, which should improve the precision of the predictions. On the other hand, the available sample is reduced from the original 767 stars to 420 stars. Therefore, there are reasons to expect an improvement in predictive performance, but also reasons to expect a degradation due to the smaller and less heterogeneous training set.

The predictions and residuals behaved similarly to those discussed in Figure 4. The characteristic GP behaviour, with a small number of objects showing extremely large predictive uncertainties, was still present, and these outliers continued to vary across different random seeds. This reinforces the interpretation that the effect is mainly related to the model and its uncertainty treatment, rather than to a fixed set of problematic stars. In this case, however, the training sample is not identical to the previous one, since only the asteroseismic stars from the original catalogue are used.

For HBNN and BART, the systematic behaviour away from $M = 1\ M_{\odot}$ was again observed, and in this case it appeared slightly amplified. The residual indicators in Table 3 support this interpretation: the absolute MRD values are larger than those reported in Table 2. This indicates that, when the models are trained only with asteroseismic stars, the predictions develop a stronger signed bias.

**Table 3.** Comparison of the relative residual metrics obtained for the different models on the holdout set. Input features are $T_{\rm eff}$, $\rho$ and [Fe/H]; target is mass.

| Metric | BART | GP | HBNN | BHS |
|---|---|---|---|---|
| MAE ($M_{\odot}$) | 0.057 | 0.053 | 0.060 | 0.052 |
| MARD [%] | 4.98 | 4.79 | 5.27 | 4.66 |
| MRD [%] | -0.63 | 0.34 | 0.71 | 0.29 |

The increase in the absolute MRD values indicates the presence of a systematic trend in the predictions after training only with asteroseismic stars. This is expected, since the training sample is now much less heterogeneous. As shown in Figure 3, the asteroseismic stars are concentrated in a relatively narrow mass range, approximately between 0.8 $M_{\odot}$ and 1.5 $M_{\odot}$. This makes the predictions more robust within this specific region of the mass spectrum, but less reliable for stars outside it. The repeatedly negative MRD values, meaning predicted masses larger than true masses, show an overprediction trend in the four models. Low-mass stars lie away from the most populated region near $\sim 1\ M_{\odot}$, and their predicted masses therefore tend to be overestimated. This behaviour is particularly interesting for the GP model: in the larger and more heterogeneous dataset no clear global trend was observed, whereas in this restricted asteroseismic sample the MRD value of $-1.32\%$ is the largest in absolute value among the four models, clearly indicating an overprediction trend.

However, the related effect of increased precision is also observed. By training the models on a specific region of mass space, the predictions become more tightly constrained within that same region. Since the holdout test is evaluated against asteroseismic stars from the same population, which mostly occupy the same region of the HR diagram, the predictions are very precise and the MARD consequently decreases. It would therefore be useful to test the same models against a different stellar sample, more evenly distributed in mass, in order to analyse whether this reduction in scatter is preserved outside the asteroseismic regime.

Regarding the BHS indicators, it is important to note that the stacking model does not yield the best values in this case. This can be understood by returning to one of the conclusions of Section 5.1: good MRD values can result from systematic mispredictions cancelling out, rather than from uniformly good predictive performance. Since an overprediction trend is now present, and since HBNN and BART previously showed stronger systematic behaviour, the stacking model is probably assigning more weight to GP over much of the analysed interval.

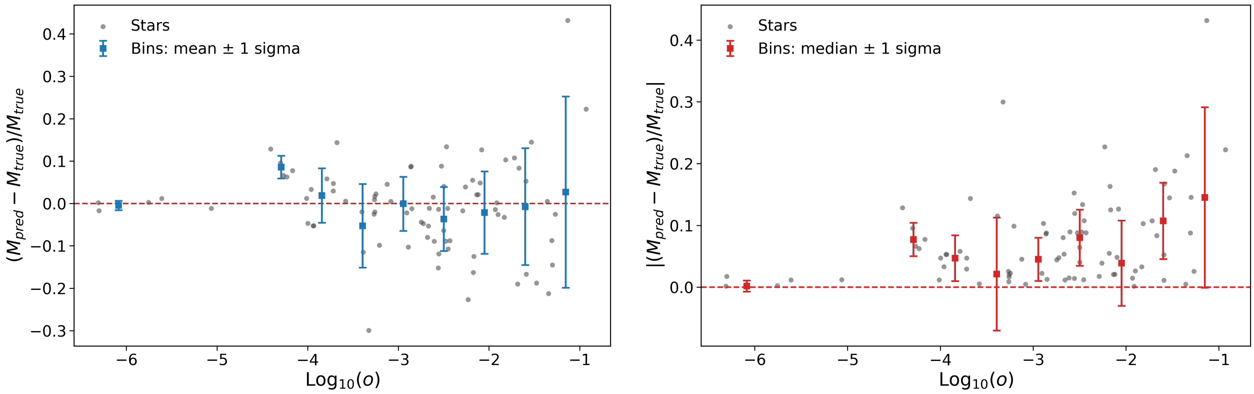
Because the holdout stars now cover a relatively narrow range of masses, if one model dominates in that region, the other two models have little opportunity to contribute in other parts of mass space. Their influence on the final stacked prediction is therefore reduced. In this run, the BHS model most likely prioritised GP, and possibly BART, over most of the analysed interval, while rarely following the HBNN predictions, which happen to give the lowest residual metrics in this particular holdout test.

### Results: oblateness

The interaction effects were studied by predicting the masses of the stars in the original dataset that belong to eclipsing-binary systems. From this group, only systems with available orbital semi-major axis $a$ could be used, since this parameter is required to compute the oblateness. As explained in Section 4.2, this leaves 295 stars with available oblateness estimates before applying the final selection cuts. Once $o$ was computed, the models trained on the asteroseismic stars were applied to these eclipsing-binary components, and the resulting mass residuals were analysed as a function of oblateness.

To avoid interpreting extrapolation effects as physical binary effects, the eclipsing-binary sample was restricted to stars well covered by the asteroseismic training set. First, only stars with $0.950 < M/M_\odot < 1.500$ were retained, corresponding approximately to the 5th and 95th mass percentiles of the asteroseismic sample. In addition, the final sample was restricted to the input-parameter domain $5500 < T_{\text{eff}} < 6800,\text{K}$, $3.750 < \log g < 4.450$, $1.000 < L/L_\odot < 11.000$ and $-0.500 < [\text{Fe/H}] < 0.270$, corresponding to the 2nd and 98th percentiles of the asteroseismic training distribution. The final eclipsing-binary sample contains $n = 85$ stars. This conservative selection reduces the number of systems, but makes the comparison more reliable, since the remaining stars lie inside the region of parameter space where the model is expected to have predictive validity.

Figure 5 shows that the oblateness values span several orders of magnitude, from $o = 4.9 \times 10^{-7}$ to $o = 1.18 \times 10^{-1}$. For this reason, the analysis was carried out using $\log o$. Globally, the signed residuals show no strong systematic tendency towards either overprediction or underprediction. The Pearson and Spearman tests also support this interpretation, since both yield correlation coefficients close to zero. The mean fractional residual is $\overline{\delta M} = -0.85\%$, while the standard deviation reaches 10.43%, driven partly by a small number of large residuals. Therefore, the model is approximately unbiased on the selected eclipsing-binary sample, but the scatter is not negligible.




**(a)** Signed fractional mass residuals.

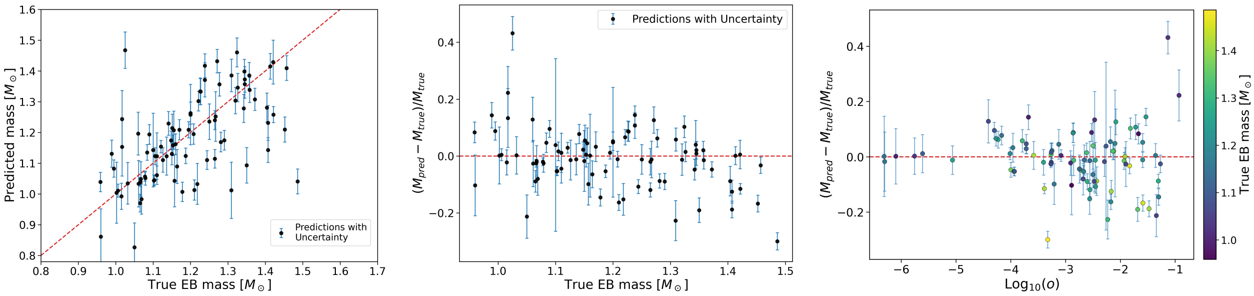
**(b)** Absolute fractional mass residuals.

**Figure 5:** Dependence of the BHS mass residuals on stellar oblateness $o$, as defined in equation 2, for the selected eclipsing-binary sample. The model uses $T_{\text{eff}}$, $L$, $\log g$ and [Fe/H] as input parameters. The left panel shows signed fractional residuals, while the right panel shows their absolute values. Grey points correspond to individual eclipsing-binary components, and the blue and red squares show binned means and medians with $1\sigma$ scatter. The sample contains $n = 85$ stars.

Figure 5b shows that the dispersion of the residuals increases with $o$. In the binned analysis, the standard deviation of $\delta M$ rises from values of a few percent at low oblateness to much larger values in the most deformed systems: 1.23%, 2.65%, 6.50%, 9.89%, 6.23%, 7.57%, 9.77%, 13.87% and 22.74%. The first bin contains only three stars and should not be overinterpreted, but the overall behaviour is consistent with a degradation of precision at larger oblateness. The same trend is visible in the absolute fractional residuals: most bins remain below about 7%, whereas the highest-oblateness bin reaches approximately 15%. Therefore, the main effect of oblateness is not a clear change in the sign of the residuals, but an increase in their spread.

This distinction is important. The results do not indicate a systematic loss of accuracy, since the average residual remains close to zero. Instead, they indicate a loss of precision: as the stars become more oblate, the model predictions become more dispersed around the true eclipsing-binary masses. This is consistent with the idea that highly distorted binary components deviate more strongly from the approximately isolated stellar evolution represented by the asteroseismic training sample. Nevertheless, this interpretation should remain cautious, because the number of stars is limited and the binned dispersions depend on the chosen bin edges.

Since Section 5.1 showed that stellar mass has a strong effect on the predictions, it is necessary to check whether the observed increase in scatter could instead be driven by $M$. Figure 6 shows that this is not the case: the residual dispersion does not display a clear dependence on stellar mass. If anything, the residual panel in Figure 6b again shows a slight tendency for the models to pull predictions towards values close to 1.25 $M_{\odot}$. This behaviour was already discussed in detail and is not associated with an increase in prediction scatter; the precision loss appears to be mass-independent. The mass-coloured residual panel in Figure 6c adds the true stellar mass as a colour scale. The points again scatter around zero, and many error bars cross the zero-residual line. At the same time, the increase in dispersion at larger oblateness remains visible, showing that the effect is not confined to a single narrow mass interval.




**(a)** Predicted masses. **(b)** Fractional residuals mass. **(c)** Fractional residuals, colour-coded by true mass.

**Figure 6:** Mass-prediction performance and mass dependence of the BHS residuals for the selected eclipsing-binary sample. The model uses $T_{\text{eff}}$, $L$, $\log g$ and [Fe/H] as input parameters. The left panel compares predicted and reference EB masses, the central panel shows the corresponding fractional residuals as a function of true mass, and the right panel shows the fractional residuals as a function of stellar oblateness $o$, colour-coded by true mass. Error bars show the propagated predictive uncertainty, and the dashed red line marks zero residual. The sample contains $n = 85$ stars.

To test the dependence of the residuals on the different input parameters, a multivariable linear regression was performed on the eclipsing-binary residuals. Each feature, together with $o$ and $M$, was assigned a regression coefficient, $\beta$, quantifying its influence on the residual behaviour. Since the predictors were standardised, the coefficients can be directly compared.

The signed residuals are not significantly associated with oblateness, with $\beta_{\log o} = 0.002 \pm 0.005$, consistent with zero. Therefore, $o$ does not introduce an overall positive or negative bias in the predictions. Instead, the signed residuals are mainly associated with the stellar parameters themselves. As already seen in Figure 6, mass has a clear effect on the tendency towards overprediction or underprediction, with $\beta_M = -0.108 \pm 0.005$. Other relevant contributions are found for $T_{\text{eff}}$, with $\beta_{T_{\text{eff}}} = 0.077 \pm 0.007$, and for [Fe/H], with $\beta_{[\text{Fe/H}]} = 0.048 \pm 0.002$.

The result changes when the absolute fractional residual is used as the response variable. In this case, $\log_{10} o$ is the only statistically significant predictor, with $\beta_{\log o} = 0.030 \pm 0.012$. Since the predictors are standardised, this means that a one-standard-deviation increase in $\log_{10} o$ is associated with an increase of about 3.0 percentage points in $|\delta M|$, after controlling for $M$, $T_{\text{eff}}$, [Fe/H], $\log g$ and $\log_{10} L$. The remaining predictors are not significant in this regression. Although the explanatory power is moderate, with $R^2 = 0.188$, this is the expected type of result: oblateness does not determine the signed residual of each individual star, but it contributes to the increase in residual magnitude, i.e. to the loss of precision.

The uncertainty calibration also suggests that eclipsing-binary stars are harder to predict than stars drawn from the training distribution, since an additional source of scatter appears when the asteroseismic-trained model is applied to eclipsing binaries. Only 30/85 stars, or 35.3%, have their true mass within $1\sigma$ of the predictive interval.

The analysis was repeated for different training seeds, keeping the same inputs, target and training strategy. The qualitative behaviour remained unchanged: the signed residuals remain approximately centred around zero, while the scatter increases with oblateness. This makes the result unlikely to be an artefact of a particular random split or training realisation.

# 6 Conclusions

This thesis developed probabilistic machine-learning models for stellar mass prediction using a curated catalogue of 767 main-sequence stars assembled mainly from detached eclipsing binaries, asteroseismology and interferometry. For FGK-type stars, using $T_{\text{eff}}$, $\rho$ and [Fe/H] as input features, the Bayesian Hierarchical Stacking model gave the best overall holdout performance, with MAE $= 0.052\,M_{\odot}$, MARD $= 4.66\%$ and MRD $= 0.29\%$. The individual models remained useful diagnostics: BART and HBNN showed a tendency to pull predictions toward the near-solar-mass region, while the GP was less biased but produced some unrealistically large predictive uncertainties.

In the ARIEL-motivated experiment, the models were trained with $T_{\text{eff}}$, [Fe/H] and $L$, avoiding $\log g$ to reduce direct dependence on stellar evolutionary models. The BHS predictions agreed most closely with the grid-based masses, reaching MAE $= 0.031\,M_{\odot}$ and MARD $= 2.86\%$, while the comparison with the linear relations of Moya & López-Sastre (2022) showed a broader and more mass-dependent residual structure. This confirms that the stacking approach is a useful data-driven benchmark, but also that averaged residual metrics must be interpreted together with residual trends and uncertainty behaviour.

The eclipsing-binary analysis shows that transferring single-star-trained models to binary components is possible, but not without limitations. For the selected sample of 85 eclipsing-binary stars, oblateness did not introduce a significant signed mass bias, with $\beta_{\log o} = 0.002 \pm 0.005$. However, it did increase the absolute residuals, with $\beta_{\log o} = 0.030 \pm 0.012$, indicating a loss of precision rather than a systematic loss of accuracy. Overall, Bayesian stacking provides competitive stellar masses and helps identify where the predictions become less reliable. The main next steps are to enlarge the calibration sample, improve GP and BHS uncertainty calibration, and repeat the oblateness test with a larger eclipsing-binary sample.

## Conclusiones

*Esta tesis desarrolló modelos probabilistas de aprendizaje automático para predecir masas estelares utilizando un catálogo depurado de 767 estrellas de secuencia principal, construido principalmente a partir de binarias eclipsantes separadas, astrosismología e interferometría. Para estrellas de tipo FGK, usando $T_{\text{eff}}$, $\rho$ y [Fe/H] como variables de entrada, el modelo de Apilamiento Bayesiano Jerárquico obtuvo el mejor rendimiento global en el conjunto de prueba, con* MAE $= 0.052\,M_{\odot}$, MARD $= 4.66\%$ *y* MRD $= 0.29\%$*. Los modelos individuales siguieron siendo diagnósticos útiles: BART y HBNN mostraron una tendencia a desplazar las predicciones hacia la región de masa casi solar, mientras que el GP presentó menos sesgo, pero produjo algunas incertidumbres predictivas no realistas.*

*En el experimento motivado por ARIEL, los modelos se entrenaron con $T_{\rm eff}$, [Fe/H] y L, evitando $\log g$ para reducir la dependencia directa de modelos de evolución estelar. Las predicciones BHS coincidieron mejor con las masas basadas en mallas evolutivas, alcanzando* $\mathrm{MAE} = 0.031\,M_\odot$ *y* $\mathrm{MARD} = 2.86\%$*, mientras que la comparación con las relaciones lineales de Moya & López-Sastre (2022) mostró una estructura residual más amplia y dependiente de la masa. Esto confirma que el apilamiento proporciona una referencia útil basada en datos, pero también que las métricas residuales medias deben interpretarse junto con las tendencias de los residuos y el comportamiento de las incertidumbres.*

*El análisis de binarias eclipsantes muestra que transferir modelos entrenados con estrellas aproximadamente aisladas a componentes binarias es posible, aunque con limitaciones. Para la muestra seleccionada de 85 estrellas binarias eclipsantes, el achatamiento no introdujo un sesgo significativo en la masa, con* $\beta_{\log o} = 0.002 \pm 0.005$*. Sin embargo, sí aumentó los residuos absolutos, con* $\beta_{\log o} = 0.030 \pm 0.012$*, lo que indica una pérdida de precisión más que una pérdida sistemática de exactitud. En conjunto, el apilamiento bayesiano proporciona masas estelares competitivas y ayuda a identificar dónde las predicciones se vuelven menos fiables. Los siguientes pasos principales son ampliar la muestra de calibración, mejorar la calibración de incertidumbres de GP y BHS, y repetir la prueba de achatamiento con una muestra mayor de binarias eclipsantes.*

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